Example 4

Simplify $x^2 + 7x - 5 + x^2 - 4x + 8$.

$x^2 + 7x - 5 + x^2 - 4x + 8 = x^2 + x^2 + 7x - 4x - 5 + 8 = 2x^2 + 3x + 3$.

Brackets

All brackets must be removed before an expression can be simplified.

Remember $4(n + 4)$ means 4 groups of "$n + 4$",

$$(n + 4) + (n + 4) + (n + 4) + (n + 4) = n + n + n + n + 4 + 4 + 4 + 4$$

i.e. 4 "n"s and 4 "4"s $= 4n + 16$.

Similarly $5(2p + q - 3) = 5(2p) + 5(q) + 5(-3)$

$$= 10p + 5q - 15$$

Example 5

Remove the brackets and simplify:

$2(3x + 2y - 4) - 3(x - 3y + 2)$

$$\begin{aligned} 2(3x + 2y - 4) - 3(x - 3y + 2) &= 6x + 4y - 8 - 3x + 9y - 6 \\ &= 6x - 3x + 4y + 9y - 8 - 6 \\ &= 3x + 13y - 14 \end{aligned}$$

Example 6

Remove the brackets and simplify the following:

$2(x^2 + x + 3) + 3(2x^2 - 3x - 1)$

$$\begin{aligned} 2(x^2 + x + 3) + 3(2x^2 - 3x - 1) &= 2x^2 + 2x + 6 + 6x^2 - 9x - 3 \\ &= 8x^2 - 7x + 3 \end{aligned}$$

Exercise 8C

Simplify each of the following by adding *like terms*.

1. $x + x + x$ **2.** $2y + y + 3y$ **3.** $2a + a + 3a + 4a$

4. $2b + 3b + 5b$ **5.** $4x - 2x + 5x$ **6.** $3a + 5a - 6a$

7. $2x + y + 4x + 6y$ **8.** $4x + 7x + 3x - 5x$ **9.** $12a + b + 3a + 4b$

10. $m + 2m - n + 3n$ **11.** $2a - b + 4b - a$ **12.** $7x + 7 - 3x - 4$

13. $6y - 2 - 3y + 8$ **14.** $5x - 4x - 3 + 7$ **15.** $x^2 + 4x + 3x^2 - 2x$

16. $3x^2 - 2x + x^2 + 3x$ **17.** $4x^2 - 3 - x^2 - 4$ **18.** $2x^2 - 3x - 5x + 4$

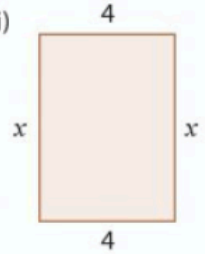
19. Multiply out the brackets of each of the following.

(a) $6(n + 1)$ (b) $5(m - 3)$ (c) $2(4 + 2k)$ (d) $5(2c + 1)$ (e) $4(3h - 5)$

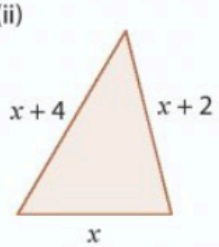
(f) $3(4a + 6)$ (g) $5(2w - 4)$ (h) $5(3 - 2p)$ (i) $6(3 - 4x)$ (j) $8(4c + 5)$

20. Find the perimeter of each of the following shapes.
Give your answers in the simplest form.

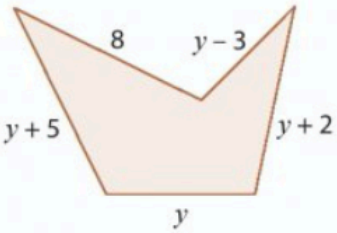
(i)



(ii)



(iii)



21. Simplify these by removing the brackets:
(i) $3(n + 6) + 4(n + 5)$ (ii) $4k + 5(4 + 3k) - 7k$

Remove the brackets in each of the following and simplify:

22. $2(x + 1) + 3(x + 3)$

23. $5(x + 4) + 2(2x + 6)$

24. $3(2x + 1) + 4(x - 3)$

25. $3(x - 6) + 2(2x + 6)$

26. $5(2x - 1) + 3(3x + 2)$

27. $2(3x - 4) + 3(3x - 5)$

28. $3(2x - 6) - 2(x + 4)$

29. $2(5x - 1) - 3(2x - 8)$

30. $5(a + b) + 3(a + 2b)$

31. $2(2a + b) + 3(4a - b)$

32. $6(2a - b) - 2(a - b)$

33. $3(2a - 3b) - 3(5a + b)$

34. $2(x^2 + 3) + 5(x^2 - 2)$

35. $3(x^2 - 1) + 2(x^2 + 4)$

36. $3(x^2 + x - 6) + 2x^2 + 6x$

37. $5(x^2 + 4x) + 2x^2 - 3x - 4$